.

EE1060: Differential Equations and Transform Techniques

Lecture 17: Second-order nonhomogeneous ODEs

Topics :

1. Method of Variation of Parameters

Method of Variation of Parameters

goal is to find $y_p(x)$ for the DE $y'' + P(x)y' + Q(x)y = g(x) \longrightarrow (1)$

$$y_p = u_1(x)\,y_1(x) + u_2(x)\,y_2(x) \longrightarrow (2)$$ where y_1 & y_2 are the LI solutions of $L[y]=0$.

Need to determine $u_1(x)$ and $u_2(x)$ S.T $\underline{\underline{y_p}}$ is a solution to $L[y]=g(x)$.

(2 conditions)

$$y_p' = u_1 y_1' + u_2 y_2' + \underbrace{(u_1' y_1 + u_2' y_2)}_{=0} \longrightarrow (3)$$ Condition (I): $u_1' y_1 + u_2' y_2 = 0$.

$$y_p'' = u_1 y_1'' + u_2 y_2'' + u_1' y_1' + u_2' y_2' \longrightarrow (4)$$

Sub (3) & (4) in (1)

$$u_1 y_1'' + u_2 y_2'' + u_1' y_1' + u_2' y_2' + P(x)\left[u_1 y_1' + u_2 y_2'\right] + Q(x)\left(u_1 y_1 + u_2 y_2\right) = g(x)$$

$$u_1\left[y_1'' + P(x)\,y_1' + Q(x)\,y_1\right] + u_2\left[y_2'' + P(x)\,y_2' + Q(x)\,y_2\right] + u_1' y_1' + u_2' y_2' = g(x)$$

Method of Variation of Parameters

$$u_1 \underbrace{\left[y_1'' + P(x)\, y_1' + Q(x)\, y_1 \right]}_{0} + u_2 \underbrace{\left[y_2'' + P(x)\, y_2' + Q(x)\, y_2 \right]}_{0} + u_1' y_1' + u_2' y_2' = g(x)$$

$$u_1' y_1' + u_2' y_2' = g(x) \longrightarrow \text{II}$$

$u_1(x)$ & $u_2(x)$:

$$\left.\begin{aligned} u_1' y_1 + u_2' y_2 &= 0 \\ u_1' y_1' + u_2' y_2' &= g(x) \end{aligned}\right\}$$

$$u_1' = -\frac{y_2\, g(x)}{y_1 y_2' - y_2 y_1'} = -\frac{y_2(x)\, g(x)}{W(y_1, y_2)}$$

$$u_2' = \frac{y_1(x)\, g(x)}{W(y_1, y_2)}$$

$$u_1 = \int u_1'\, dx = \int -\frac{y_2(x)\, g(x)}{W(y_1, y_2)}\, dx \quad \text{and} \quad u_2 = \int \frac{y_1(x)\, g(x)}{W(y_1, y_2)}\, dx.$$

$$y_p(x) = u_1(x)\, y_1(x) + u_2(x)\, y_2(x).$$

Method of Variation of Parameters

goal: find $y = y_c + y_p$ for $L[y] = g(x)$

Variation of Parameters:

① Solve for $y_c \longrightarrow (y_1, y_2)$

② $y_p = u_1(x)\, y_1(x) + u_2(x)\, y_2(x)$

where $u_1 = \int \dfrac{-y_2(x)\, g(x)}{W(y_1, y_2)}\, dx.$

$u_2(x) = \int \dfrac{y_1(x)\, g(x)}{W(y_1, y_2)}\, dx.$

③ $y = y_c + y_p.$

Undetermined Coefficients:

① Compute $\phi(D)$ for $g(x)$

② Solve $\phi(D)\, P(D)\, y = 0.$

$$y = \sum_{i=1}^{n} c_i y_i + \underbrace{\sum_{i=n+1}^{n+m} c_i y_i}_{\text{I}} \longrightarrow$$

find c_i such that I is the particular solution.

Example

find the Complete solution of ODE $\frac{d^2y}{dx^2} + k^2 y = \sin \omega x$ when $\omega \neq k^2$.

$$y_c = c_1 \cos(kx) + c_2 \sin(kx) \quad (\text{or}) \quad A\cos(kx+\phi) \quad (\text{or}) \; A\underline{\underline{e^{j(kx+\phi)}}}$$

$$y_1 = \cos kx$$

$$y_2 = \sin kx.$$

$$y_p = \underline{\underline{v_1(x)}} \cos kx + \underline{\underline{v_2(x)}} \sin kx.$$

$$W(y_1, y_2) = \begin{vmatrix} \cos kx & \sin kx \\ -k\sin kx & k\cos kx \end{vmatrix} = k.$$

$$v_1(x) = -\frac{1}{k}\int \sin(kx)\sin(\omega x)\, dx.$$

$$= -\frac{1}{2k}\left[\frac{\sin(k-\omega)x}{k-\omega} - \frac{\sin(k+\omega)x}{k+\omega}\right]$$

$$v_2(x) = \frac{1}{k}\int \cos(kx)\sin(\omega x)\, dx = \frac{-1}{2k}\left[\frac{\cos(k-\omega)x}{k-\omega} + \frac{\cos(k+\omega)x}{k+\omega}\right]$$

$$y_p(x) = v_1(x)\cos kx + v_2 \sin kx.$$